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Properties of Materials
Lecture 13

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Lecture 9: Recap from Last Time, Task
Exercise Seminar 2 / Lecture 10: Packing of Atoms in Solids (1), Learning Objectives, The Structure of Metals and Ceramics (1), Unit Cell Parameters, Face-Centred Cubic (fcc), Hexagonal Close-Packed (hcp), Crystal Structures - Metals, Crystal Structures - Ceramics, Concepts and Terminology (5)
Lecture 11: Packing of Atoms in Solids (2), Learning Objectives, Atomic Packing, Atomic Packing - bcc Structures, Atomic Packing Factor - bcc, Atomic Packing - fcc Structures, Atomic Packing Factor - fcc, Atomic Packing - hcp Structures, Atomic Packing Factor - hcp, Tips for Calculating APF for hcp, Concepts and Terminology (6)
Lecture 12: The Physical Basis of Young's Modulus, Learning Objectives, Moduli of Crystals, Rubbers and the Glass Transition Temperature, Composite, Composite Terminology, Fibre-Reinforced Composites, Rule of Mixtures (1)
Lecture 13: Rule of Mixtures (2), Task, Concepts and Terminology (7)

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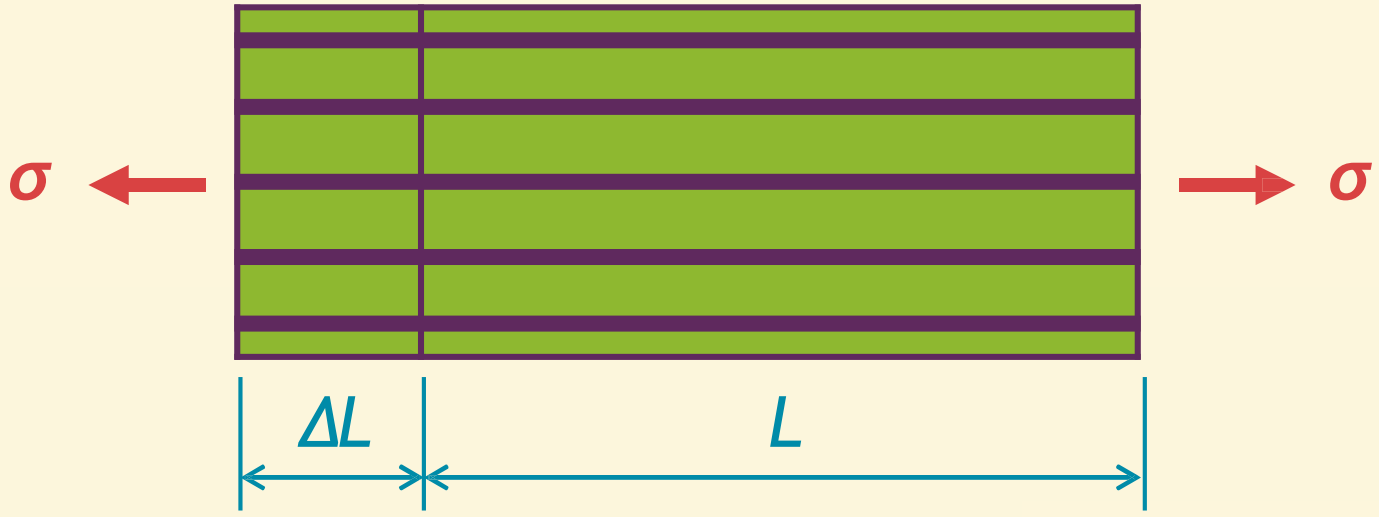
The Physical Basis of Young's Modulus

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Rule of Mixtures

Rule of mixtures (1)

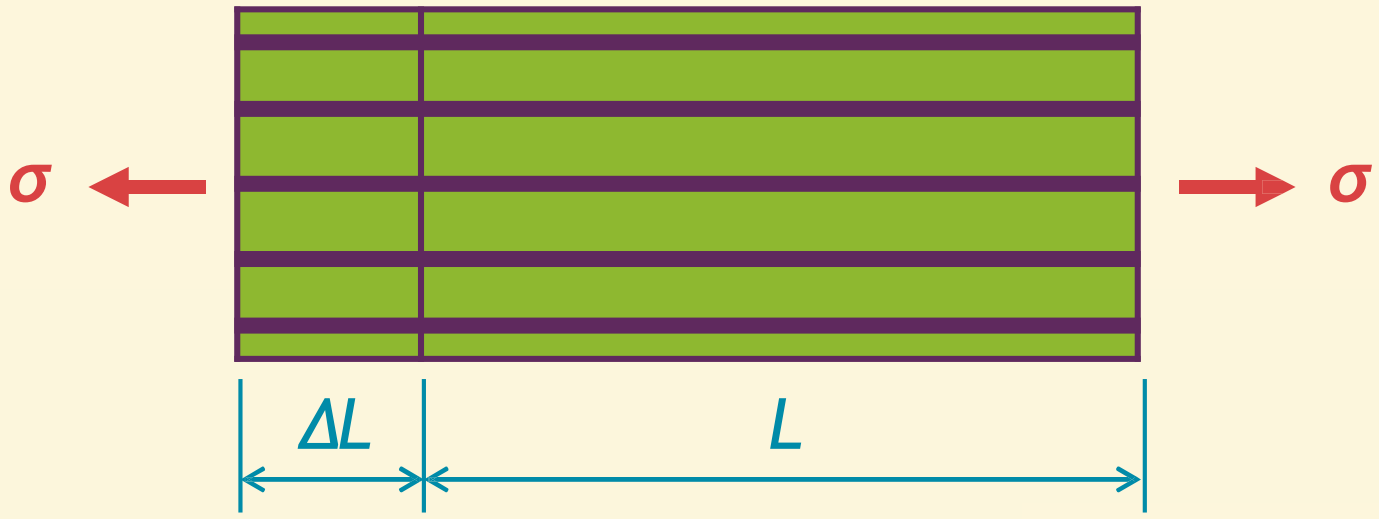
- Estimating the modulus of composites **(upper limit)**:
  - Subject composite to a known stress, **parallel** to the fibre direction
  - As fibres and matrix are bonded, they stretch by the same amount –e.g they have the **same strain, ε_n**
  - Assuming **equal proportions** of fibres and matrix, $\sigma = \left(\sigma_f + \sigma_m\right)/2$



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Rule of mixtures (2)

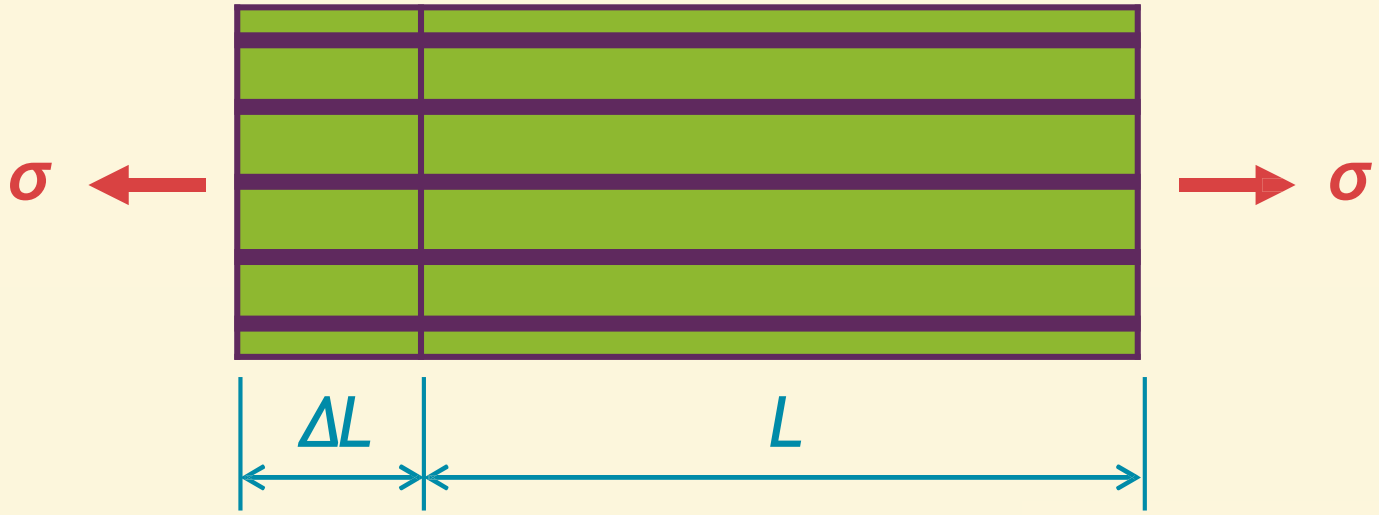
- In equal proportions, $\sigma = \left(\sigma_f + \sigma_m \right)/2$
- In un-equal proportions, $\sigma = V_f \sigma_f + \left(1 - V_f \right) \sigma_m$
  - V_f = volume fraction of fibres
- Since $\sigma = E \varepsilon_n$:
  - $\sigma = \left[E_f V_f + E_m \left(1 - V_f \right) \right] \varepsilon_n$



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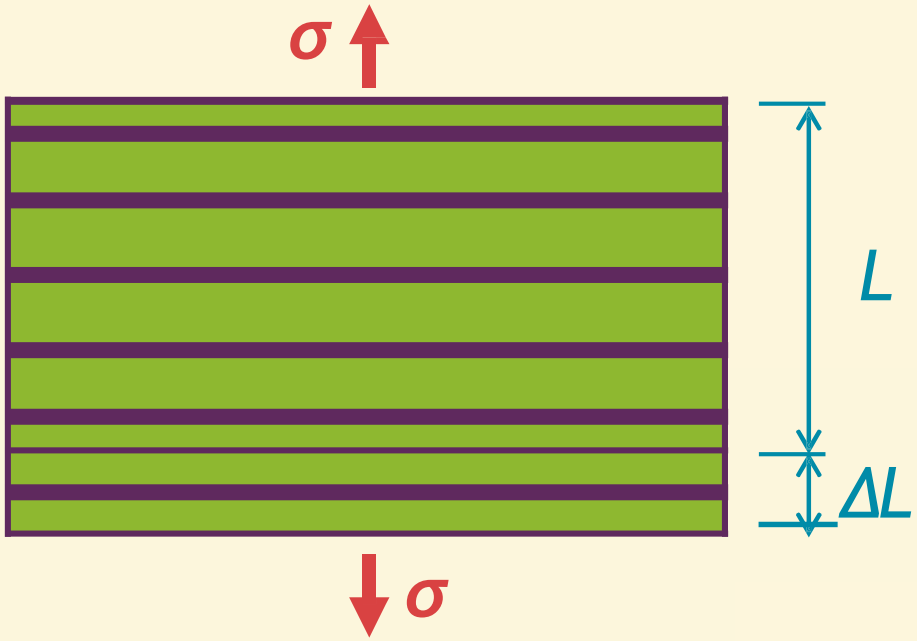
Rule of mixtures (3)

- $\sigma = \left[E_f V_f + E_m\left(1 - V_f\right)\right]\varepsilon_n$
- As $E_{\text{composite}} = \frac{\sigma}{\varepsilon_n}$:
  - $E_{\text{composite}} = E_f V_f + E_m\left(1 - V_f\right)$
- **Upper estimate** for the composite modulus:
  - Need to consider loading perpendicular to fibres



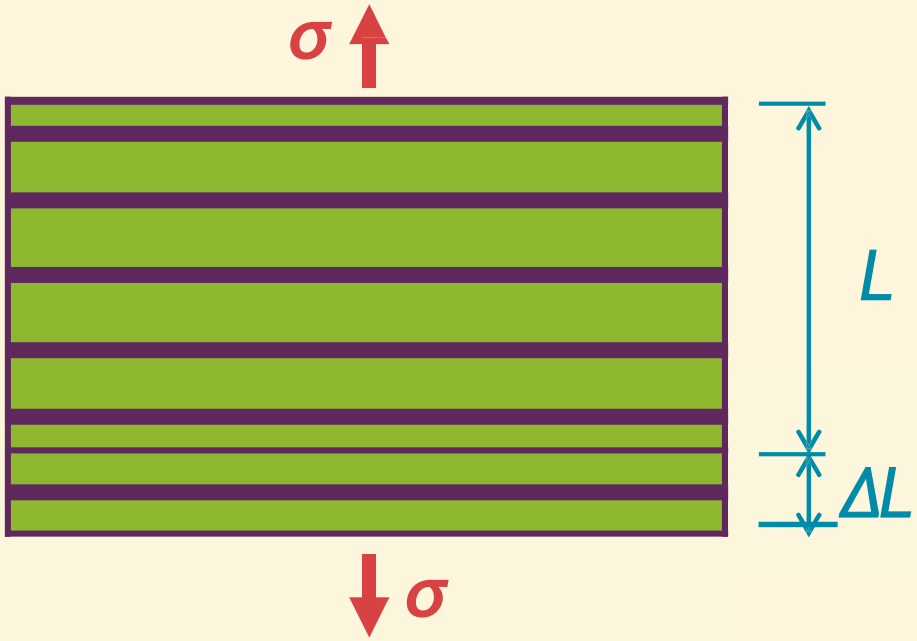
Rule of mixtures (4)

- Estimating the modulus of composites **(lower limit)**:
  - Subject composite to a known stress, **perpendicular** to the fibre direction
  - In this direction, it is reasonable to assume that they share the **same stress** and therefore have a weighted nominal strain
    - In **equal proportions**, $\varepsilon_n = (\varepsilon_{n,f} + \varepsilon_{n,m})/2$



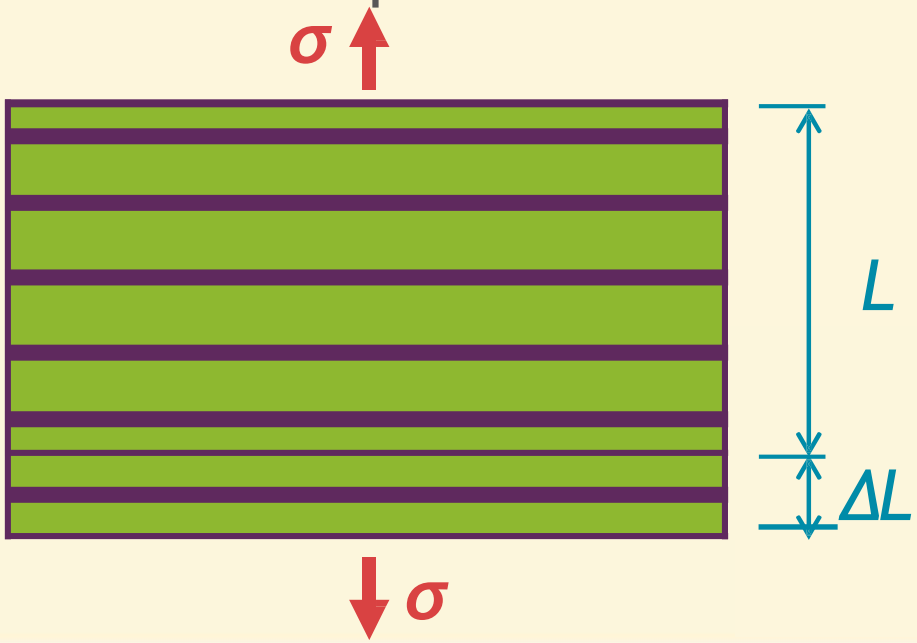
Rule of mixtures (5)

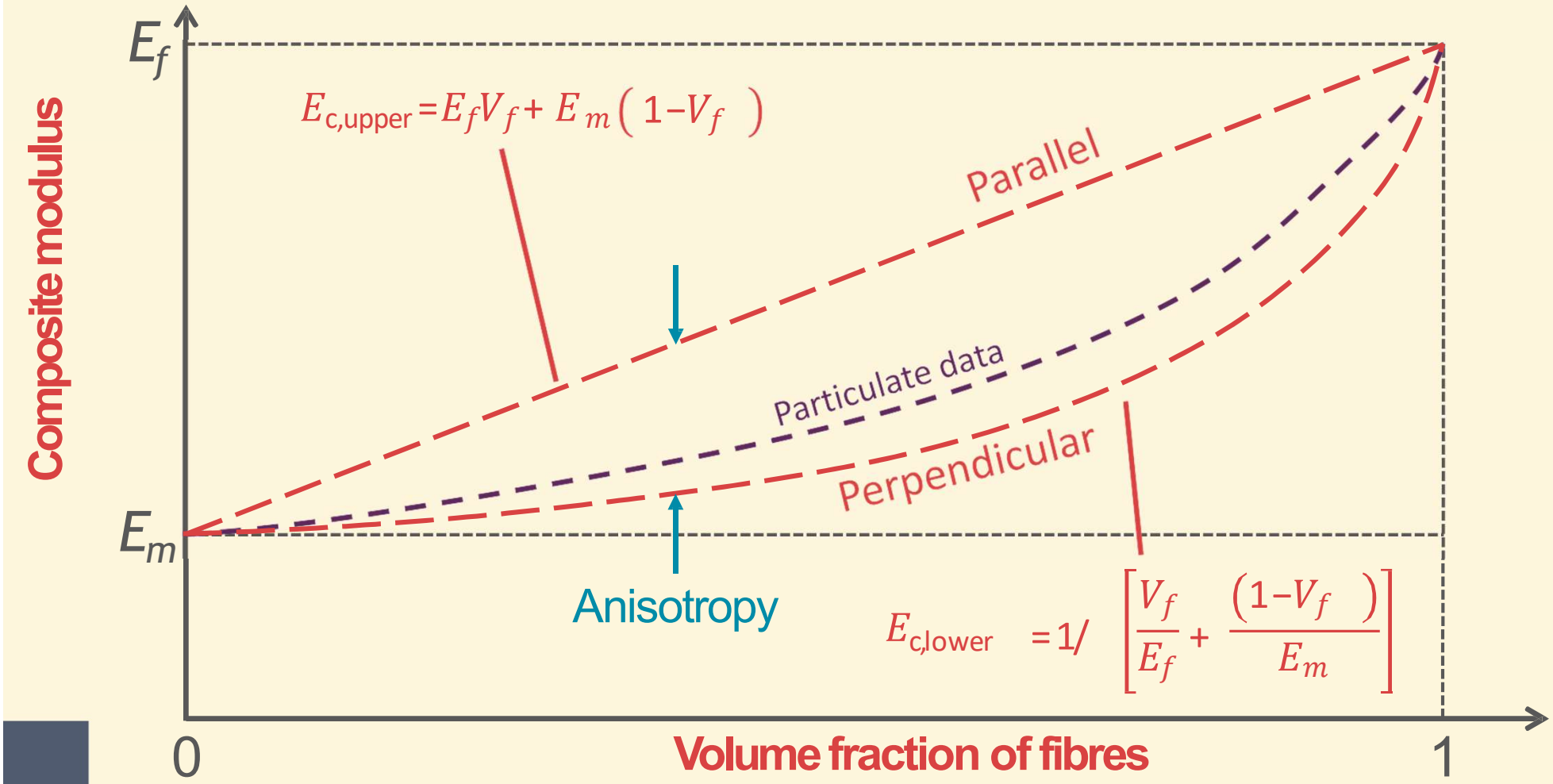
- In equal proportions, $\varepsilon_n = \left(\varepsilon_{n,f} + \varepsilon_{n,m} \right)/2$
- In un-equal proportions, $\varepsilon_n = V_f \varepsilon_{n,f} + \left(1 - V_f\right) \varepsilon_{n,m}$
- Since $\sigma = E \varepsilon_n$:
  - $\varepsilon_n = \left[\frac{V_f}{E_f} + \frac{\left(1 - V_f\right)}{E_m} \right] \sigma$



Rule of mixtures (6)

- $\varepsilon_n = \left[\frac{V_f}{E_f} + \frac{(1-V_f)}{E_m}\right]\sigma$
- As $E_{\text{composite}} = \frac{\sigma}{\varepsilon_n}$:
  - $E_{\text{composite}} = 1/\left[\frac{V_f}{E_f} + \frac{(1-V_f)}{E_m}\right]$
- **Lower estimate** for the composite modulus





Rule of mixtures (7)

- Fibre-reinforced composites are **anisotropic**:
  - Different strength depending on loading direction
- Particulate composites are **isotropic**

Rule of mixtures (8)

- **Alignment** 'efficiency':
  - Fibre orientation with respect to loading axis
  - **Warning:** Transverse properties can be *worse* than un-reinforced matrix!
  - We include **orientation factor**, η_O
    e.g. $E_c = \eta_O E_f V_f + E_m (1 - V_f)$
  - (2D random, $\eta_O = 3/8$; 3D random, $\eta_O = 1/5$)

- Overcoming anisotropy:
  - **Laminating** so alternate layers have different fibre directions

Summary

- The moduli of metals, ceramics and glassy polymers below T_G reflect the **stiffness** of the bonds which link the atoms.
- Glasses and glassy polymers above T_G are leathers, rubbers or viscous liquids, and have much **lower moduli**.
- **Composites** have moduli which are a weighted average of those of their components.

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Concepts and terminology

- Stress-strain curves:
  - **Brittle** and **ductile** materials
- **Stiffness**:
  - Hooke's Law
  - Young's modulus
- **Linear elasticity**
- Engineering / **nominal** stress and strain
- **True** stress and strain